

AI-POWERED REAL ESTATE INTELLIGENCE, VALUATION & INVESTMENT RECOMMENDATION PLATFORM

Final Documentation Report

Project Report Submitted for Internship Project Submission

Submitted By

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Internship Role: Intern (AI/ML)

Project Title: AI-Powered Real Estate Intelligence, Valuation & Investment Recommendation Platform

Technology Stack:

Python | Machine Learning | Scikit-Learn | XGBoost | CatBoost | LightGBM | Pandas | NumPy | Streamlit | Plotly | SHAP | ML Models

Project Description:

An AI-powered real estate intelligence system developed to predict property prices, estimate future growth, and provide intelligent investment recommendations. The project uses machine learning models for price prediction, SHAP for explainability, and risk analytics for buy/hold/sell decisions. A Streamlit dashboard is developed to visualize insights and support data-driven real estate investments.

Submitted To

Company/Organization Name: Data Vidwan

Mentor/Manager Name: Meet Mistry

Date of Submission: 25/06/2026

AI-POWERED REAL ESTATE INTELLIGENCE, VALUATION & INVESTMENT RECOMMENDATION PLATFORM

1. Project Objective

The objective of this project is to develop an AI-driven real estate decision support system that helps businesses and investors evaluate property prices, forecast future growth, and make intelligent investment decisions.

The system uses machine learning-based regression algorithms along with investment analytics and risk optimization techniques to identify undervalued properties, reduce investment risk, and support better financial decisions.

The project includes:

- Exploratory Data Analysis
- Data preprocessing
- Feature engineering
- Price prediction model development
- Model comparison
- Risk scoring and assessment
- Investment recommendation intelligence
- Explainable AI (SHAP)
- Interactive Streamlit dashboard

2. Requirement Coverage

Requirement	Status	Project File / Output
Price Prediction Model	Completed	models/catboost_log_model.pkl
Feature Engineering	Completed	notebooks/
Cleaned Dataset	Completed	data/processed/house_price_final.csv
Price Forecast Output	Completed	data/processed/house_price_with_risk_recommendation.csv
Risk Optimization	Completed	data/processed/house_price_with_risk.csv
Investment Recommendation	Completed	data/processed/house_price_with_risk_recommendation.csv

EDA Report	Completed	reports/figures/
Model Evaluation	Completed	notebooks/
Streamlit Dashboard	Completed	dashboard/app.py
Project Documentation	Completed	reports/Internship_Documentation_Report.docx

3. Dataset

The project uses comprehensive real estate datasets:

Real Estate Property Dataset

Records: 15,000+

Features:

- Property Type (Apartment/Villa)
- BHK & Bathrooms
- Area (sq.ft) & Property Age
- Locality, City, and Pincode
- Floor Number & Furnished Status
- Proximity Features (Airport, School distance)
- Infrastructure Score & Condition
- Price (Target Variable)

4. Work Completed

Completed an end-to-end AI/ML workflow including:

- Loaded and inspected real estate datasets.
- Performed data quality validation.
- Handled missing values and outlier detection.
- Performed exploratory data analysis.
- Generated business-focused visualizations.
- Created advanced property valuation features.
- Trained multiple machine learning regression models.
- Compared model performance (CatBoost, XGBoost, Random Forest, etc.).
- Selected the best performing model (CatBoost).
- Generated property price predictions.

- Implemented property risk analysis engine.
- Calculated 1Y, 3Y, 5Y future forecasts.
- Generated investment recommendations (Buy/Hold/Sell).
- Implemented Explainable AI using SHAP.
- Developed fully functional Streamlit dashboard.
- Exported processed datasets and trained model.

5. Methodology

5.1 Data Validation

Dataset quality was analyzed by checking:

- Missing values
- Duplicate records
- Data types
- Statistical distributions

No duplicate records were identified or they were handled appropriately.

5.2 Exploratory Data Analysis

Performed analysis on:

Real Estate Dataset:

- Price distribution
- Area vs Price distribution
- Location/Pincode distribution
- Correlation Matrix
- BHK vs Price heatmaps

5.3 Feature Engineering

Created additional analytical features:

Valuation Features:

- Log Transformation of Target (Price)
- Infrastructure Score
- Market Multiplier

Risk & Recommendation Features:

- Risk Score Calculation
- Investment Score
- Buy/Hold/Sell Logic

6. Model Development

Multiple regression models were trained and compared:

Models evaluated:

- Linear Regression
- Random Forest Regressor
- XGBoost Regressor
- Gradient Boosting Regressor
- LightGBM
- CatBoost Regressor

The models were evaluated using:

- MAE
- RMSE
- R^2 Score

7. Model Results

Model	MAE	RMSE	R^2 Score
CatBoost	60713.90	103548.36	0.9239
LightGBM	61886.93	105655.59	0.9207
XGBoost	61453.95	109080.93	0.9155
Gradient Boosting	64571.56	110417.36	0.9134
Random Forest	68158.58	129923.30	0.8802
Linear Regression	106893.83	196359.26	0.7263

8. Selected Model

CatBoost Regressor

Reason for selection:

After evaluating multiple regression algorithms, CatBoost provided the best overall performance with the lowest error values and highest R^2 score among tested models. It successfully handled the categorical variance without requiring heavy one-hot encoding.

Final Model:

models/catboost_log_model.pkl

Performance:

Metric	Score
MAE	60713.90
RMSE	103548.36
R ² Score	0.9239

9. Streamlit Application

The Streamlit application is implemented in:

dashboard/app.py

The dashboard contains:

Executive Dashboard

Displays:

- Project Overview
- EDA Dashboard (Histograms, Heatmaps)

Property Price Prediction Page

Features:

- User Inputs for Area, BHK, Location, Property Age
- Outputs Predicted Price

Investment & Risk Analysis Pages

Features:

- Investment Recommendation (Buy/Hold/Sell)
- Risk Score and Risk Category

Future Forecasting & Explainability Pages

Features:

- 1-Year, 3-Year, 5-Year Forecasts
- SHAP Summary Plot & Feature Importance

Market Insights

Allows users to inspect Top Investment Areas, High Growth Areas, and Low Risk Areas.

10. Key Findings

- Identify undervalued properties using accurate CatBoost forecasts.
- Assess long-term growth through 5-year ROI projections.
- Monitor properties with high risk scores based on age and low infrastructure.
- Adjust investment strategies based on locality premiums.
- Use Explainable AI (SHAP) to justify valuation to stakeholders.

10. Recommendations

- Target 'BUY' recommendations in low-risk, high-ROI areas.
- Investigate properties showing significant undervaluation.
- Prioritize comprehensive due diligence for 'High Risk' orders.

11. Achievements

Successfully completed:

- Data preprocessing & Exploratory analysis
- Machine learning model development with high R^2 (0.9239)
- Property price forecasting
- Investment & Risk optimization engine
- SHAP XAI implementation
- End-to-End Streamlit Dashboard deployment

12. Limitations

- Dataset contains limited macroeconomic indicators.
- Price prediction performance depends on historical locality records.
- Real-time API feeds for live real estate listings are not implemented.
- Recommendations are rule-based rather than dynamic reinforcement learning.

13. Final Submission Contents

- Documentation Report
- Machine Learning Models (.pkl)
- Streamlit Dashboard (app.py)
- Processed Datasets
- EDA Visualizations
- Jupyter Notebooks
- Presentation Slides (.pptx)

14. How to Run

```
pip install -r requirements.txt
```

```
streamlit run dashboard/app.py
```

15. Final Conclusion

The AI-Powered Real Estate Intelligence, Valuation & Investment Recommendation Platform successfully delivers an end-to-end machine learning solution for price forecasting, risk monitoring, and investment optimization.

The system combines CatBoost predictions with robust business analytics and Explainable AI (SHAP) to help investors reduce risks, identify lucrative opportunities, and make data-driven decisions through a fully interactive Streamlit dashboard.